

Chapter 5

Production Theory

"Production is not the application of tools to materials, but logic to work."

— Peter Drucker, *The Practice of Management*, 1954

Connecting Weeks 1–4 to Week 5

In Weeks 1–4 you built a complete theory of the *consumer*: how preferences are represented by indifference curves, how optimal choices are found at the budget-line tangency, how demand curves are derived, and how elasticity and consumer surplus measure responsiveness and welfare. **This week we build the parallel theory for the *firm*.** Every consumer concept has an exact production-theory counterpart: the utility function becomes the production function, indifference curves become isoquants, the MRS becomes the MRTS. Learning these parallels efficiently is the key to mastering production theory.

Questions This Chapter Will Answer

- Why does adding the tenth worker to a factory add less output than the second worker?
- How can a firm produce the same output using more labour and less capital — or vice versa?
- What does it mean for a technology to be “efficient”?
- Why can a software company double its output by hiring fewer than twice as many engineers?
- Why does a traditional craft brewery struggle to double output even when it doubles all inputs?
- What is the production-theory analogue of every consumer-theory concept you already know?

Learning Objectives and Topics

Each objective states what you will be able to *do*. Sections listed beneath.

LO1 Recall the consumer theory framework and identify its production-theory counterpart for every concept.

§5.1 *Consumer–Production Analogy*

LO2 Define the production function and explain what “maximum output” means; distinguish production from utility.

§5.2 *Production Function* §5.3 *Total Product*

LO3 Compute marginal product, explain the law of diminishing MP, and relate TP and MP curves.

§5.4 *Marginal Product*

LO4 Draw and interpret isoquants; identify the three special cases (smooth convex, linear, L-shaped).

§5.5 *Isoquants*

LO5 Compute the MRTS, explain diminishing MRTS, and distinguish input substitution from technological change.

§5.6 *MRTS* §5.7 *Technological Change*

LO6 Classify a production function as CRS, IRS, or DRS and apply the Cobb-Douglas diagnostic ($a + b$).

§5.8 *Returns to Scale* §5.9 *Worked Examples*

Notation Used in This Chapter.

Symbol	Meaning	Symbol	Meaning
Q	output quantity	MP_L	marginal product of labour
K	capital input	MP_K	marginal product of capital
L	labour input	$MRTS_{L,K}$	marginal rate of technical substitution
$f(K, L)$	production function	λ	scale factor ($\lambda > 1$ = scaling up)
$TP(L)$	total product (K fixed)	a, b	output elasticities in Cobb-Douglas
A	technology parameter (Cobb-Douglas)	CRS/IRS/DRS	returns to scale classification

Before You Read — Warm-Up Question

Maria runs a small food-truck business. She currently employs two cooks and owns one grill. She is thinking about hiring a third cook.

Q1. Do you expect the third cook to be more productive, equally productive, or less productive than the second cook? Why?

Q2. If Maria buys a second grill *at the same time* as hiring the third cook, does your answer change? Why?

Keep your intuition in mind as you read. The concepts of marginal product (§5.4) and returns to scale (§5.8) will give you the formal tools to answer both questions precisely.

5.1. The Consumer–Production Analogy: A Compact Review

You have spent four weeks mastering consumer theory. How much of that work carries over to production theory — and exactly how do the two frameworks mirror each other?

Rather than re-deriving consumer theory from scratch, we present the complete mapping in a compact two-panel table. Every concept from Weeks 1–4 has an exact production-theory counterpart. Learning the new material is largely a matter of recognising these parallels.

The Consumer–Production Analogy: Complete Mapping

Consumer Theory (Weeks 1–4)		Production Theory (Week 5)	
Utility function	$U(x, y)$	Production function	$Q = f(K, L)$
Inputs to utility	Goods x and y	Inputs to output	Capital K , Labour L
Utility level	U_0 (subjective)	Output level	Q_0 (objective, measurable)
Equal-utility curve	Indifference curve	Equal-output curve	Isoquant
Slope of IC	$-MRS_{xy}$	Slope of IQ	$-MRTS_{LK}$
$MRS = MU_x / MU_y$	trade-off rate	$MRTS = MP_L / MP_K$	substitution rate
Diminishing MRS	convex IC	Diminishing MRTS	convex isoquant
IC shifts outward	higher utility	IQ shifts <i>inward</i>	technological improvement
Optimal choice	$MRS = P_x / P_y$	Cost-min. choice	$MRTS = w / r$ (Week 6)
Budget constraint	$P_x x + P_y y = I$	Isocost line	$wL + rK = C$ (Week 6)
CS / welfare measure	consumer surplus	Producer surplus	firm profit (later)

Critical difference: Utility is **subjective and unobservable**. Output is **objective and measurable**. A firm can count its output; a consumer cannot count their utility.

The Core Analogy in One Sentence. Just as consumers *maximise utility* subject to a budget constraint, firms *minimise cost* subject to a production target. The isoquant is the firm's indifference curve; the isocost line ($wL + rK = C$) is the firm's budget line; and the optimality condition $MRTS = w/r$ is the firm's version of $MRS = P_x/P_y$. Every consumer-theory result you already know has an exact production-theory mirror.

Case Study: A Bakery. The bakery's production function answers: given a certain number of bakers (labour L) and ovens (capital K), what is the *maximum* number of loaves that can be produced per week? Notice the word “maximum” — if workers are idle or ovens misused, actual output falls below the production function. This is exactly like the consumer's utility function: it gives maximum utility from a bundle, assuming the consumer behaves optimally.

Check Your Understanding

1. Match each consumer concept to its production counterpart: (a) indifference curve; (b) MRS; (c) “budget line pivots outward”; (d) shift of IC.
2. Why does a technological improvement shift the isoquant *inward* rather than outward (as a higher IC would)? What is the economic logic?
3. In one sentence: what is the fundamental difference between utility and output that makes production theory more empirically tractable?

With the analogy firmly in place, we now build the production framework from scratch — starting with the production function itself.

5.2. The Production Function

A bakery has five ovens and ten bakers. What is the maximum number of loaves it can produce per week? The production function answers this question — and pins down exactly how inputs combine to create output.

Intuition Before the Formula. Think of the production function as a recipe book for the firm. Just as a recipe specifies the maximum amount of dish you can make from given ingredients when the chef is doing their best, the production function specifies the maximum output from given inputs when they are used as efficiently as possible.

Production Function: A production function specifies the **maximum** output the firm can produce for any combination of inputs. With capital K and labour L :

$$Q = f(K, L) \quad (5.1)$$

In words: The production function maps any input bundle (K, L) to the highest possible output level — assuming inputs are used as efficiently as possible.

The phrase *maximum* is important: the production function describes the *technological frontier*. Actual output can be lower if inputs are wasted or mismanaged, but cannot exceed the function's value.

$Q = f(K, L)$: the firm's recipe for turning capital and labour into output.

We simplify to two inputs (K and L) for analytical tractability. In practice, firms combine many inputs — raw materials, energy, managerial talent. The two-input framework captures all the essential economics and the logic extends directly to many inputs.

Maria's Food Truck. Maria runs a food truck selling tacos. Her production function is $Q = f(K, L)$, where Q is the number of tacos produced per hour, K is the number of grills (capital), and L is the number of cooks (labour). With one grill and two cooks, she can produce at most 40 tacos per hour. Notice “at most”: if one cook is idle or the grill is not running at full temperature, actual output is lower. Maria's production function describes her technological frontier — the best she can do, not the average of what she does.

We will follow Maria throughout this chapter as we build the full toolkit: how her output changes when she hires more cooks (§5.3–5.4), what her isoquant looks like (§5.5–5.6), and whether doubling her truck size would double her output (§5.8).

Why This Matters. Production theory is the foundation for understanding firm behaviour: why firms hire workers, invest in equipment, and adopt new technology. Every business decision about scaling up, automating, or relocating traces back to the production function. Understanding its structure helps managers predict how costs change as output grows, and helps policymakers design subsidies or regulations that affect production decisions.

Check Your Understanding

1. Why does the production function give the *maximum* output rather than actual output? Give an example of a situation where actual output would fall below the production function.
2. For $Q = K^{0.5}L^{0.5}$: compute output at $(K = 4, L = 4)$, $(K = 1, L = 9)$, and

($K = 9, L = 1$). Verified: $Q = 4, 3, 3$ respectively?

3. True or false: “Two firms with the same inputs must produce the same output.” Explain.

The production function gives output for any (K, L) combination. To understand how output changes when inputs change, we study total product and marginal product.

5.3. Total Product

Suppose a factory has a fixed number of machines. As we hire more and more workers, does output keep rising at the same rate — or does it eventually slow down?

Total Product (TP): When capital is fixed at $\bar{K}$ and only labour varies, the resulting relationship between L and output is the **total product curve**:

$$TP(L) = f(\bar{K}, L) \quad (5.2)$$

In words: Total product is simply the output you get from L units of labour, given a fixed amount of capital. It traces out what happens to output as you add more workers.

Numerical Example: $Q = 10L - 0.5L^2$ (with K fixed)

Using the production relationship $TP = 10L - 0.5L^2$ (with capital fixed), compute total product as labour increases:

L (workers)	TP (units/week)	MP_L	Phase
0	0	—	—
1	9.5	9.5	Rising fast
2	18.0	8.5	Rising fast
4	32.0	7.0	Slowing
6	42.0	5.0	Slowing
8	48.0	3.0	Slowing
10	50.0	1.0	Near peak
11	49.5	−0.5	Declining

Total product *peaks* at $L = 10$ (where MP_L first turns zero) and *declines* beyond that. No rational firm would hire the 11th worker.

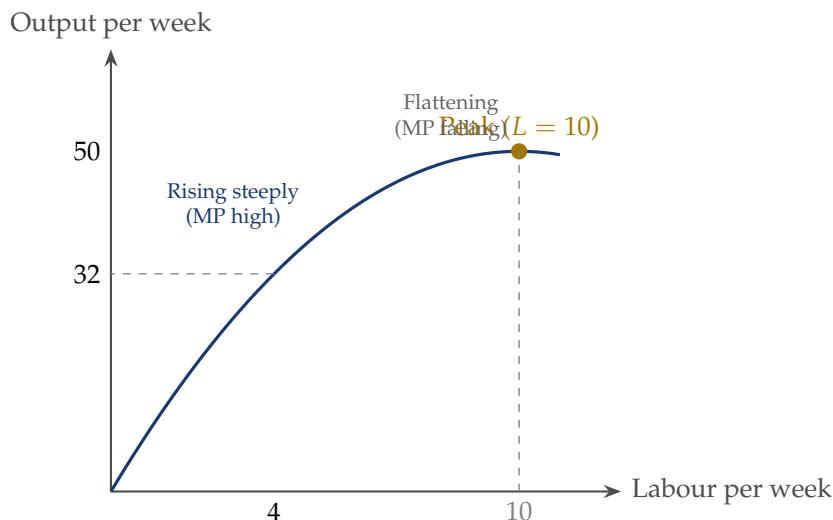


Figure 5.1. Total product curve for $TP = 10L - 0.5L^2$. What to look for: TP rises steeply at first (when each worker has ample capital to work with), then flattens as diminishing returns set in. TP peaks at $L = 10$ where $MP_L = 0$, then declines. A rational firm stops hiring before the peak.

Shape of the Total Product Curve

Check Your Understanding

1. For $TP = 10L - 0.5L^2$: at what value of L does TP peak? Verify by setting $MP_L = dTP/dL = 0$.
2. Why would no rational firm hire the 11th worker (where $MP_L < 0$)? What does it mean for total product when $MP_L < 0$?
3. A student says: "The TP curve always slopes upward." True or false? Explain.

Total product shows the overall output level. Marginal product tells us how much each additional worker adds to that output — and why those additions get smaller and smaller.

5.4. Marginal Product

The tenth worker adds only 1 unit of output per week while the first added 9.5. Why does each additional worker contribute less and less as the workforce grows?

5.4.1. Definition and Formula

Intuition Before the Formula. Marginal product answers: if I hire one more worker (holding everything else fixed), how much extra output do I get? When workers are scarce, each new hire is very productive — they have plenty of machines to work with. As the workforce grows, machines become congested, and each additional worker adds less. This is the law of diminishing marginal product in a nutshell.

Marginal Product of Labour (MP_L): The additional output from one more unit of labour, holding capital fixed:

$$MP_L = \frac{\Delta TP}{\Delta L} \Big|_{K \text{ fixed}} = \frac{\partial f(K, L)}{\partial L} \quad (5.3)$$

Similarly, the **marginal product of capital** is:

$$MP_K = \frac{\Delta TP}{\Delta K} \Big|_{L \text{ fixed}} = \frac{\partial f(K, L)}{\partial K} \quad (5.4)$$

In words: MP_L = extra output per extra worker (capital fixed). MP_K = extra output per extra machine (labour fixed).

“The first cook makes tacos; the fifth cook makes excuses.” This is the law of diminishing marginal product — not a joke about cooks, but a precise statement about what happens when you add workers to a fixed kitchen.

$MP_L = \partial Q / \partial L$:
the production
counterpart of
marginal utility
 $MU_x = \partial U / \partial x$.

5.4.2. The Law of Diminishing Marginal Product

Law of Diminishing Marginal Product: As the quantity of one input increases (other inputs held fixed), its marginal product eventually **declines**. This is not a technological assumption — it follows from the fact that additional workers must share a fixed stock of capital. As the worker-to-machine ratio rises, each worker is less and less well-equipped.

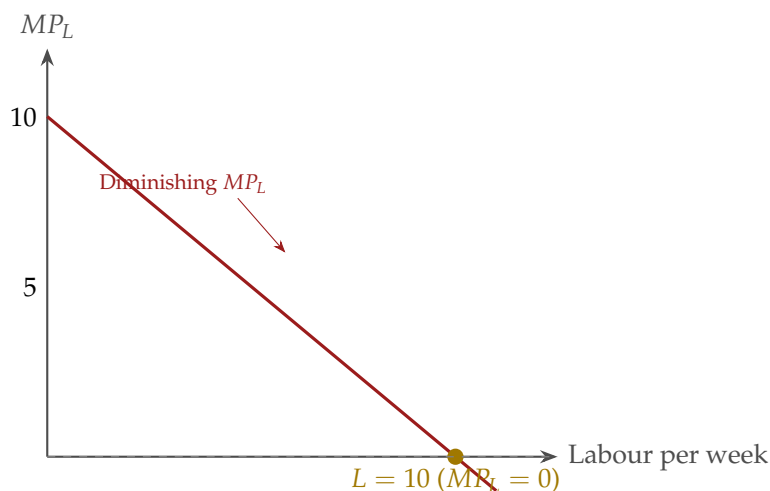


Figure 5.2. Marginal product of labour for $MP_L = 10 - L$. What to look for: MP_L starts at 10 (the first worker is highly productive) and falls linearly. At $L = 10$, $MP_L = 0$ — the point where total product peaks. Hiring beyond $L = 10$ would give negative MP_L , reducing total output.

5.4.3. Relationship Between TP and MP

The TP and MP curves are linked: MP_L is the *slope* of the TP curve at each point.

TP–MP Relationship:

- Where TP is **rising steeply**: MP_L is **high and positive**.

- Where TP is **rising slowly** (flattening): MP_L is **small and positive**.
- Where TP reaches its **peak**: $MP_L = 0$.
- Where TP is **declining**: $MP_L < 0$.

Worked Example: MP from Cobb-Douglas.

For $Q = K^{0.5}L^{0.5}$, compute MP_L and MP_K analytically.

$$MP_L = \frac{\partial Q}{\partial L} = 0.5 K^{0.5} L^{-0.5} = \frac{1}{2} \sqrt{\frac{K}{L}}$$

$$MP_K = \frac{\partial Q}{\partial K} = 0.5 K^{-0.5} L^{0.5} = \frac{1}{2} \sqrt{\frac{L}{K}}$$

In words: Each input's MP is proportional to the square root of the other input divided by itself — capturing the idea that a worker is more productive when surrounded by more machines.

Verify diminishing MP_L : As L increases (holding K fixed), $L^{-0.5}$ falls, so MP_L falls. ✓

Evaluate at ($K = 4, L = 1$): $MP_L = 0.5\sqrt{4/1} = 1.0$; $MP_K = 0.5\sqrt{1/4} = 0.25$.

Evaluate at ($K = 4, L = 4$): $MP_L = 0.5\sqrt{4/4} = 0.5$; $MP_K = 0.5\sqrt{4/4} = 0.5$.

As L rises from 1 to 4 (with $K = 4$ fixed), MP_L falls from 1.0 to 0.5 — confirming diminishing marginal product.

Maria's food truck ($Q = K^{0.5}L^{0.5}$, $K = 1$ fixed): At $L = 1$, $MP_L = 0.5$. At $L = 4$, $MP_L = 0.25$. At $L = 9$, $MP_L = 0.17$. Each additional cook adds tacos, but progressively fewer — the single grill is the binding constraint.

Economic Insight. Diminishing marginal product applies when only *one* input changes. This is a statement about what happens when one factor is constrained — not a statement about overall technology. A firm can simultaneously exhibit diminishing MP_L and increasing returns to scale (as we will see in Section 5.8).

Application: Diminishing Returns in Real Production.

Manufacturing: A car assembly plant with 20 robots can run efficiently with 40 workers. Adding the 41st worker may add little because the bottleneck is robot capacity, not labour. MP_L is diminishing at this margin.

Agriculture: Applying fertiliser to a field follows a classic diminishing-returns pattern. The first tonne of fertiliser dramatically raises crop yield; the tenth tonne adds very little because the limiting factor has become soil quality or water.

Knowledge work: Adding the tenth programmer to a project of nine people may slow things down (coordination costs) — MP_L can even turn negative, exactly as in our numerical table.

Historical Episode: The US Manufacturing Productivity Slowdown, 1970s.

Between 1948 and 1973, US manufacturing output per worker grew at roughly 2.8% per year. After 1973, growth slowed sharply — to under 1% per year through much of the decade. One important (though not the only) explanation is diminishing marginal product operating at the industry level.

During the postwar boom, manufacturing capital — factories, machinery, assembly lines — expanded rapidly and labour productivity was high because each worker commanded substantial capital. By the early 1970s, the capital stock in many sectors had grown faster than labour-augmenting technology could absorb it, and additional investment in some industries yielded falling incremental output per dollar invested. When the 1973 oil shock simultaneously raised energy costs (a complementary input), marginal products fell further.

Lesson for theory: Even at the macro level, MP_L is not constant. When capital grows faster than complementary improvements in technology or management, diminishing returns to the fixed factor (here, absorptive capacity) set in. Returns to scale — the question we turn to in §5.8 — addresses what happens when *all* inputs grow together, which is a fundamentally different question.

Source: Gordon (2016), *The Rise and Fall of American Growth*; Baily & Gordon (1988).

Try It. For $Q = K^{0.5}L^{0.5}$ with $K = 9$ fixed: compute MP_L at $L = 1$, $L = 4$, and $L = 9$. Confirm that MP_L is diminishing. What is Q at each point?

Check Your Understanding

1. For $TP = 10L - 0.5L^2$: use calculus to find MP_L . At what L does $MP_L = 0$? Consistent with the table?
2. “Diminishing marginal product means the firm is becoming less efficient.” True or false? Explain carefully.
3. For $Q = K^{0.5}L^{0.5}$: as K increases (holding L fixed), does MP_L rise or fall? What does this tell you about how more capital affects workers’ productivity?

We now understand how output responds when one input changes while the other is fixed. The next step is to look at all combinations of both inputs that produce the same output level — the isoquant.

5.5. Isoquants

A factory can produce 100 units using 4 machines and 4 workers, or using 1 machine and 16 workers. Both combinations are equally productive. How do we represent all such equivalent combinations in a single diagram?

Intuition Before the Definition. An isoquant is to the firm exactly what an indifference curve is to the consumer. Just as an indifference curve shows all bundles giving equal utility, an isoquant shows all input combinations giving equal output. The prefix “iso” means “same” — an isoquant is a “same quantity” curve.

Isoquant: A curve in (L, K) space showing all input combinations that produce a given output level Q_0 :

$$\text{Isoquant for } Q_0 : \{ (K, L) : f(K, L) = Q_0 \} \quad (5.5)$$

In words: An isoquant traces out the menu of ways to achieve a given output target — using more labour and less capital, or vice versa.

5.5.1. Standard Isoquant: $Q = K^{0.5}L^{0.5}$

For our benchmark production function $Q = K^{0.5}L^{0.5}$, the isoquant for output level Q_0 satisfies $K^{0.5}L^{0.5} = Q_0$, giving $K = Q_0^2/L$.

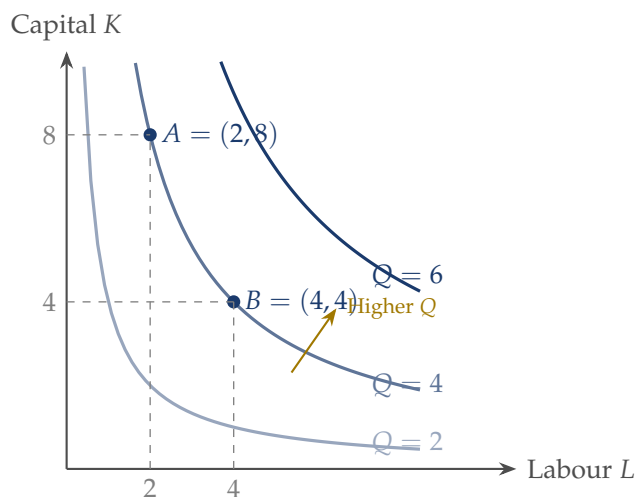


Figure 5.3. Isoquant map for $Q = K^{0.5}L^{0.5}$: three output levels. What to look for: Points $A = (L = 2, K = 8)$ and $B = (L = 4, K = 4)$ both lie on the $Q = 4$ isoquant — verify: $\sqrt{8 \times 2} = 4 \checkmark$ and $\sqrt{4 \times 4} = 4 \checkmark$. Higher isoquants (further from origin) represent higher output. Isoquants never cross.

5.5.2. Three Special Cases of Isoquants

Just as indifference curves have three special cases (standard, perfect substitutes, perfect complements), isoquants do too:

“Which Isoquant Is This?” Decision Guide

Shape	Type	Production function	Intuition
Smooth, convex	Standard	$Q = AK^aL^b$ (Cobb-Douglas)	Labour and capital are imperfect substitutes
Straight line	Perfect substitutes	$Q = aK + bL$ (linear)	Capital and labour are interchangeable
L-shaped (kink)	Perfect complements	$Q = \min(aK, bL)$ (Leontief)	Fixed proportions required

Note: For linear production, $MRTS$ is constant (straight line). For Leontief, $MRTS$ is 0 or ∞ except at the kink. For Cobb-Douglas, $MRTS$ diminishes smoothly.

In plain English:
 Smooth convex isoquant $\Leftrightarrow$ imperfect substitutes.
 Straight line $\Leftrightarrow$ perfect substitutes.

Check Your Understanding

1. For $Q = K^{0.5}L^{0.5}$: find two input combinations on the isoquant $Q = 6$. Verify both satisfy $K^{0.5}L^{0.5} = 6$.
2. A robot and a human worker can both bolt car doors at the same rate, and can be used interchangeably. What shape would the isoquant for car doors bolted per hour look like?
3. True or false: “Two isoquants can cross if they represent very different output levels.” Use the analogy with indifference curves to justify your answer.

Isoquants show all the ways to produce a given output. The slope of an isoquant tells us the rate at which the firm can trade one input for another while keeping output constant — that is the MRTS.

5.6. The Marginal Rate of Technical Substitution (MRTS)

If a factory hires one more worker, how many machines can it remove while keeping output unchanged? The MRTS answers this question — and explains why the answer changes as the factory becomes more labour-intensive.

5.6.1. Definition and Formula

Intuition Before the Formula. The MRTS is to the firm exactly what the MRS is to the consumer. The MRS asked: how many units of Y will the consumer give up for one more unit of X, staying equally happy? The MRTS asks: how many units of capital can the firm give up for one more unit of labour, keeping output constant? Both measure a trade-off rate along an equal-value curve.

Marginal Rate of Technical Substitution ($MRTS_{L,K}$): The rate at which capital can be reduced per unit increase in labour, holding output constant:

$$MRTS_{L,K} = - \left. \frac{\Delta K}{\Delta L} \right|_{Q \text{ fixed}} = \frac{MP_L}{MP_K} \quad (5.6)$$

In words: $MRTS = \text{extra output from one more worker} \div \text{extra output from one more machine} = \text{how many machines one worker can replace}.$

The formula $MRTS = MP_L / MP_K$ has an elegant intuition: if one extra worker produces MP_L units, and one extra machine produces MP_K units, then to keep output constant after adding a worker you can remove MP_L / MP_K machines.

*$MRTS = MP_L / MP_K$:
exactly mirrors
 $MRS = MU_x / MU_y$ from
consumer theory.*

5.6.2. Worked Example: MRTS for $Q = K^{0.5}L^{0.5}$

$MP_L = 0.5K^{0.5}L^{-0.5}$ and $MP_K = 0.5K^{-0.5}L^{0.5}$ (from Section 5.4).

$$MRTS = \frac{MP_L}{MP_K} = \frac{0.5K^{0.5}L^{-0.5}}{0.5K^{-0.5}L^{0.5}} = \frac{K}{L}$$

Evaluate at key points on the $Q = 4$ isoquant:

(L, K)	$Q = K^{0.5}L^{0.5}$	$MRTS = K/L$	Interpretation
(1, 16)	4 ✓	16	1 worker replaces 16 machines
(2, 8)	4 ✓	4	1 worker replaces 4 machines
(4, 4)	4 ✓	1	1 worker replaces 1 machine
(8, 2)	4 ✓	0.25	1 worker replaces $\frac{1}{4}$ machine
(16, 1)	4 ✓	0.0625	1 worker replaces $\frac{1}{16}$ machine

In words: As the firm uses more labour (moving right along the isoquant), each worker becomes less productive relative to capital, so each worker can only replace a smaller and smaller amount of capital.

Economic Insight. The MRTS falls from 16 to 0.0625 as we move along the $Q = 4$ isoquant from capital-intensive (1, 16) to labour-intensive (16, 1). This is **diminishing MRTS** — the production analogue of diminishing MRS. It is why isoquants bow inward toward the origin, just as indifference curves do.

5.6.3. Diminishing MRTS

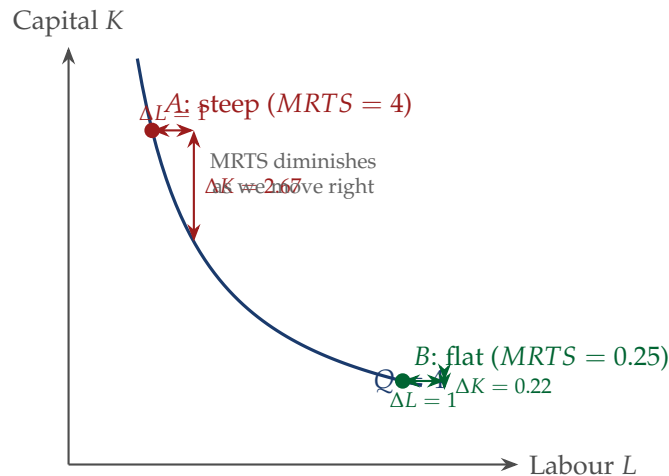


Figure 5.4. Diminishing MRTS along $Q = 4$ isoquant. At $A = (L = 2, K = 8)$: one extra unit of labour replaces 2.67 units of capital ($MRTS = K/L = 4$). At $B = (L = 8, K = 2)$: one extra unit of labour replaces only 0.22 units of capital ($MRTS = 0.25$). All values verified for $Q = K^{0.5}L^{0.5}$.

Application: Robots vs. Workers — MRTS in Manufacturing.

A car assembly plant must weld 10,000 car frames per week. It can achieve this using 50 robots and 100 workers, or 30 robots and 180 workers. At the capital-intensive point (50 robots, 100 workers), each additional worker is very productive (the firm has lots of robots but relatively few people to operate them), so the MRTS is high — each worker can replace several robots. As the firm shifts to using more workers and fewer robots, the MRTS falls: each additional worker adds less because the bottleneck is now robot capacity, not labour. This is exactly diminishing MRTS in action.

Policy implication: When governments subsidise automation (lowering the effective price of robots), firms move up their isoquants toward capital-intensive combinations where the MRTS is high. This can raise output per worker substantially.

Check Your Understanding

1. For $Q = K^{0.5}L^{0.5}$: verify that $MRTS = K/L = 9$ at $(K = 9, L = 1)$ and $MRTS = 0.11$ at $(K = 1, L = 9)$. Both points are on the $Q = 3$ isoquant.
2. A firm's $MRTS = 3$ at its current input combination. If it hires one more worker, how much capital can it remove while keeping output constant?
3. "A straight-line isoquant implies constant MRTS." True or false? What type of production function generates a straight-line isoquant?

We now understand how the MRTS describes substitution possibilities along an isoquant. But what happens when technology itself changes — not just the input mix?

5.7. Technological Change and the Isoquant

A factory adopts a new automated assembly line. Is this the same as hiring more workers — or something fundamentally different? How do we distinguish changing the input mix from changing the technology itself?

Two distinct phenomena — easy to confuse:

- **Input substitution:** changing the *mix* of K and L to produce the same output. Movement **along** the isoquant. Technology unchanged — the isoquant does not move.
- **Technological change:** the production *technology itself* improves. The same output is now achievable with fewer resources. The isoquant **shifts inward** (toward the origin).

Common Mistake: Buying machines $\neq$ technological change. If a firm simply replaces some workers with machines using *existing* technology, that is input substitution — a movement along the isoquant. Technological change occurs only when the production process itself improves, so that the same output can be achieved with *fewer total resources*. The isoquant shifts only when technology changes.

In plain English:
Input substitution = move along isoquant. Tech change = shift of isoquant inward.

Check Your Understanding

1. A garment factory switches from hand-stitching (labour-intensive) to semi-automated sewing (more capital). Is this input substitution or technological change? Explain.
2. The same factory then adopts a new robotic system that cuts total input use by 30%. Now is it input substitution or technological change?
3. Why does a technological improvement shift the isoquant *inward* rather than outward? How does this differ from how better technology affects a consumer's

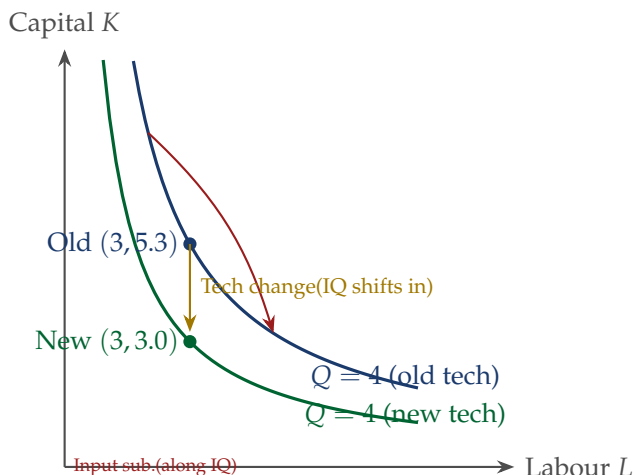


Figure 5.5. Input substitution vs. technological change. *What to look for:* The red arrow shows input substitution — moving along the same isoquant, the firm uses more L and less K but stays on the same technology. The gold arrow shows technological change — the isoquant for $Q = 4$ shifts inward, so the same output is now achievable with fewer inputs at every combination.

indifference curve?

We have seen how the firm's isoquant can shift with technology. Now we ask the most fundamental question about scale: if the firm doubles all its inputs, does output double, more than double, or less than double?

5.8. Returns to Scale

A technology firm doubles its workforce and office space. Does software output double? More than double? Less than double? The answer determines whether being a large firm is advantageous — and it shapes the structure of entire industries.

Intuition Before the Definition. Diminishing marginal product describes what happens when you increase *one* input while holding others fixed. Returns to scale describes something completely different: what happens when you increase *all* inputs proportionally. A firm can simultaneously have diminishing MP_L and exhibit increasing returns to scale — these are not contradictory.

Returns to Scale: Compares the proportional change in output to the proportional change in *all* inputs. If all inputs are scaled by factor $\lambda > 1$:

$$\text{Compare: } f(\lambda K, \lambda L) \quad \text{vs.} \quad \lambda \cdot f(K, L) \quad (5.7)$$

In words: Returns to scale asks: when all inputs scale up by the same factor, does output scale up by more, less, or the same factor?

“Two chefs, two grills, and twice the kitchen — that is the returns-to-scale question. Just one extra chef squeezed into the same kitchen — that is diminishing marginal product.” The distinction is simple but it is the most common source of confusion in this chapter.

5.8.1. Three Cases: A Summary Table

Returns to Scale: Three-Case Summary

Type	Formal condition	What it means	Real example
CRS	$f(\lambda K, \lambda L) = \lambda f(K, L)$	Double inputs $\Rightarrow$ double output	Agriculture (broadly)
IRS	$f(\lambda K, \lambda L) > \lambda f(K, L)$	Double inputs $\Rightarrow$ more than double	Software, networks
DRS	$f(\lambda K, \lambda L) < \lambda f(K, L)$	Double inputs $\Rightarrow$ less than double	Artisanal craft, mining

Cobb-Douglas diagnostic: For $Q = AK^aL^b$: CRS if $a + b = 1$; IRS if $a + b > 1$; DRS if $a + b < 1$.

5.8.2. Constant Returns to Scale (CRS)

CRS: $f(\lambda K, \lambda L) = \lambda \cdot f(K, L)$ for all $\lambda > 1$. Inputs double $\Rightarrow$ output doubles.

Isoquant signature: Isoquants are *equally spaced* along any ray from the origin.

Example: $Q = K^{0.5}L^{0.5}$ has $a + b = 1$, so CRS. Verify:

$$f(\lambda K, \lambda L) = (\lambda K)^{0.5}(\lambda L)^{0.5} = \lambda^{0.5} \cdot \lambda^{0.5} \cdot K^{0.5}L^{0.5} = \lambda \cdot Q \quad \checkmark$$

5.8.3. Increasing Returns to Scale (IRS)

IRS: $f(\lambda K, \lambda L) > \lambda \cdot f(K, L)$ for all $\lambda > 1$. Inputs double $\Rightarrow$ output more than doubles.

Isoquant signature: Isoquants become *more closely spaced* as output increases.

Application: Returns to Scale in Real Industries.

Software / digital platforms (IRS): Writing the codebase requires large up-front costs, but once built, the platform can serve additional users at near-zero marginal cost. Doubling engineers more than doubles what can be built because team members specialise, share tools, and reuse code modules. Amazon, Google, and Meta all benefit from IRS — it is a primary driver of winner-take-most dynamics in technology markets.

Agriculture (roughly CRS): A farm that doubles its land, labour, and equipment roughly doubles output. There is little specialisation advantage from being large, which is why agricultural markets support many similarly-sized farms.

Artisanal / craft production (DRS): An artisanal cheesemaker who doubles workers and aging caves does not double output — managing a larger operation introduces coordination problems and quality-control difficulties. Very large artisanal operations lose the precision that defines their product.

The Economist's Lens: Semiconductors and Increasing Returns

The concept in practice: IRS and the CHIPS Act.

Few industries illustrate increasing returns to scale more vividly than semiconductor fabrication. Building a cutting-edge chip fabrication plant (a “fab”) costs between \$10 billion and \$20 billion. Once built, the marginal cost of producing an additional wafer is relatively modest — overwhelmingly, the cost is the plant itself. Double the number of wafers produced and average cost falls sharply, because that enormous fixed cost is spread over more units. This IRS structure has profound consequences. Empirical estimates of output elasticities in semiconductor manufacturing suggest that a 1% increase in all inputs is associated with approximately a 1.1–1.3% increase in output — implying $a + b \approx 1.1\text{--}1.3$ in a Cobb-Douglas representation (estimates from Hall, 1988; Basu & Fernald, 1997; and more recently Syverson, 2011). In plain language: the industry exhibits meaningful IRS, meaning larger fabs are substantially cheaper per chip than smaller ones.

This is precisely why global chip production has consolidated into a handful of firms (TSMC, Samsung, Intel) and a small number of geographic clusters (Taiwan, South Korea, Arizona). A new entrant cannot match the per-unit cost of an incumbent running a fully utilised, depreciated fab — the IRS structure creates a natural barrier to entry.

Policy application: The US CHIPS and Science Act (2022) committed \$52 billion in subsidies partly to shift the equilibrium — to make it financially viable to build new fabs on US soil despite these IRS-driven concentration pressures. From a production-theory standpoint, the subsidy is designed to help new US fabs reach the scale at which their average cost becomes competitive. Returns to scale, in other words, is not just classroom theory: it is the explicit economic logic behind a major piece of industrial policy.

Source: Hall (1988, JPE); Basu & Fernald (1997, JPE); Syverson (2011, JEL); Congressional Budget Office (2022).

5.8.4. Decreasing Returns to Scale (DRS)

DRS: $f(\lambda K, \lambda L) < \lambda \cdot f(K, L)$ for all $\lambda > 1$. Inputs double $\Rightarrow$ output less than doubles.

Isoquant signature: Isoquants become *more widely spaced* as output increases.

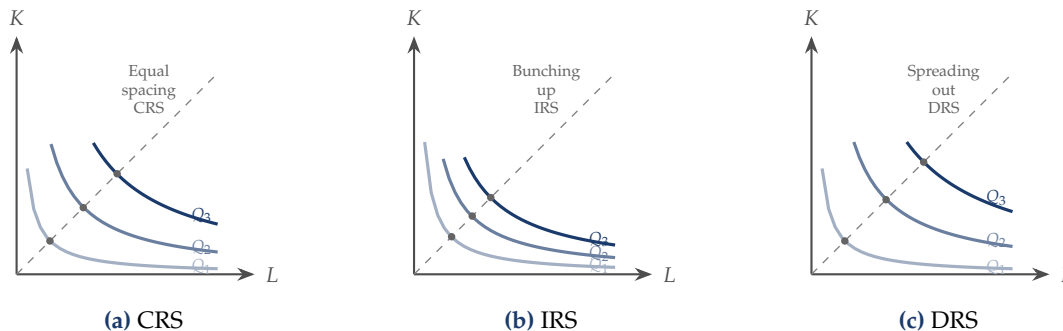


Figure 5.6. Three returns-to-scale cases along a ray from the origin. What to look for: For CRS, the three grey dots (equal proportional input increases) are equally spaced along the ray. For IRS, the dots bunch up (output doubles before inputs double). For DRS, the dots spread out (need more than double inputs to double output).

Maria revisited. Maria currently operates one grill ($K = 1$) and two cooks ($L = 2$). If she doubles her food truck — two grills, four cooks, twice the counter space — does she produce exactly twice as many tacos? The answer depends on her production function's returns to scale. If $Q = K^{0.5}L^{0.5}$ (CRS): at $(1, 2)$, $Q = \sqrt{2} \approx 1.41$; at $(2, 4)$, $Q = \sqrt{8} \approx 2.83 = 2 \times 1.41$. Yes — exactly twice. CRS means Maria gets no *size advantage* (or disadvantage) from scaling up. A second food truck is simply as efficient as the first.

Check Your Understanding

1. For $Q = K^{0.6}L^{0.6}$: compute Q at $(K = 1, L = 1)$ and at $(K = 2, L = 2)$. Does output more than double? Confirm using the Cobb-Douglas diagnostic.
2. A government wants to encourage competition in an industry with IRS. Why is this harder than in a CRS industry? What market structure tends to emerge naturally from IRS?
3. For $Q = K^{0.3}L^{0.3}$: classify as CRS, IRS, or DRS. If inputs double from $(K = 1, L = 1)$, by what factor does output increase?

We have now covered all the core concepts. The next section brings them together through three fully worked numerical examples.

5.9. Worked Examples

How do we apply the production function, MRTS, and returns-to-scale test all at once? This section works through three canonical production functions step by step, showing every calculation.

5.9.1. Example 1: Linear Production Function ($Q = 2K + L$)

Production function: $Q = 2K + L$ (frisbee factory).

(a) Find the input combination to produce $Q = 100$ with $K = 10$:

$$100 = 2(10) + L \implies L = 80$$

(b) Find the input combination to produce $Q = 100$ with $K = 25$:

$$100 = 2(25) + L \implies L = 50$$

(c) Isoquant for $Q = 100$: Rearranging: $K = 50 - \frac{1}{2}L$. This is a straight line with K -intercept 50 and L -intercept 100.

(d) What type of isoquant? Straight line $\implies$ **perfect substitutes**. Capital and labour are interchangeable at a constant rate of 2:1 (one unit of capital always replaces two units of labour).

In words: The linear production function says capital is always twice as productive as labour. You can always swap one machine for two workers, regardless of how many you already have.

(e) MRTS:

$$MP_L = \frac{\partial Q}{\partial L} = 1, \quad MP_K = \frac{\partial Q}{\partial K} = 2$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{1}{2}$$

MRTS is **constant** (not diminishing) because the isoquant is a straight line.

(f) Returns to scale:

$$f(\lambda K, \lambda L) = 2(\lambda K) + \lambda L = \lambda(2K + L) = \lambda Q$$

This is **CRS** ($a + b = 1$ if rewritten as $K^1 L^0$ style, but directly verified above).

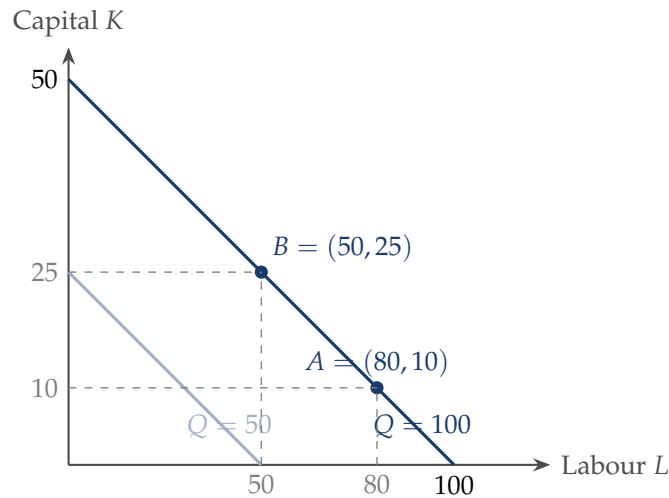


Figure 5.7. Linear isoquants for $Q = 2K + L$. What to look for: Straight-line isoquants confirm perfect substitution between capital and labour. Points $A = (L = 80, K = 10)$ and $B = (L = 50, K = 25)$ both satisfy $2(10) + 80 = 100 \checkmark$ and $2(25) + 50 = 100 \checkmark$. The constant slope ($-\frac{1}{2}$) confirms constant $MRTS = \frac{1}{2}$.

5.9.2. Example 2: Cobb-Douglas ($Q = K^{0.5}L^{0.5}$)

Production function: $Q = K^{0.5}L^{0.5}$ (our benchmark throughout this chapter).

(a) Marginal products:

$$MP_L = 0.5K^{0.5}L^{-0.5} = \frac{K^{0.5}}{2L^{0.5}}, \quad MP_K = 0.5K^{-0.5}L^{0.5} = \frac{L^{0.5}}{2K^{0.5}}$$

(b) MRTS:

$$MRTS = \frac{MP_L}{MP_K} = \frac{K^{0.5}/2L^{0.5}}{L^{0.5}/2K^{0.5}} = \frac{K}{L}$$

In words: The MRTS equals the capital-to-labour ratio. When the firm is capital-intensive (high K/L), each worker can replace many machines. When labour-intensive (low K/L), each worker can replace very few machines.

(c) Verify diminishing MRTS along the $Q = 4$ isoquant ($K = 16/L$):

At $(L = 2, K = 8)$: $MRTS = 8/2 = 4$. At $(L = 4, K = 4)$: $MRTS = 4/4 = 1$. At $(L = 8, K = 2)$: $MRTS = 2/8 = 0.25$. ✓ Declining.

(d) Returns to scale:

$$f(\lambda K, \lambda L) = (\lambda K)^{0.5}(\lambda L)^{0.5} = \lambda^{0.5+0.5}K^{0.5}L^{0.5} = \lambda^1 Q$$

Since $a + b = 0.5 + 0.5 = 1$: **CRS**. ✓

(e) Numerical check: At $(K = 2, L = 2)$: $Q = 2$. At $(K = 4, L = 4)$: $Q = 4 = 2 \times 2$. ✓ CRS confirmed.

Economic Insight. For any Cobb-Douglas $Q = AK^aL^b$, the sum $a + b$ is a one-number diagnostic for returns to scale. It is also the *scale elasticity* — the percentage increase in output when all inputs rise by 1%. This elegant property makes Cobb-Douglas the most widely estimated production function in empirical economics.

5.9.3. Example 3: Identifying Returns to Scale from the Exponents

True or False: “Every production function with diminishing MP_L must have decreasing returns to scale.”

Answer: FALSE.

Consider $Q = K^{0.6}L^{0.6}$:

- $MP_L = 0.6K^{0.6}L^{-0.4}$. As L rises (holding K fixed), $L^{-0.4}$ falls $\Rightarrow MP_L$ falls. **Diminishing MP_L** . ✓
- $a + b = 0.6 + 0.6 = 1.2 > 1 \Rightarrow$ **IRS**.

At $(K = 1, L = 1)$: $Q = 1$. At $(K = 2, L = 2)$: $Q = 2^{0.6} \times 2^{0.6} = 2^{1.2} \approx 2.30$. Output more than doubled. ✓

In words: Diminishing marginal product is about what happens when you vary one input. Returns to scale is about what happens when you vary all inputs proportionally. These are completely separate questions and can go in opposite directions.

The worked examples demonstrate the mechanics. Before turning to the practice questions, we collect the exam mistakes, back-matter reference material, and revision summary.

The worked examples above have shown the full toolkit for analysing a production function: compute marginal products, derive MRTS, identify returns to scale, and sketch isoquants. The practice questions that follow apply all of these skills — both in forward problems (given the production function, answer questions about it) and reverse problems (given observed input-output data, identify the underlying production technology). Attempt every question before consulting the solutions.

Where Does This Take Us?

- **Week 6** introduces the firm’s cost-minimisation problem: given input prices w (wage) and r (rental rate), the firm chooses the least-cost input combination to produce a given output. The optimality condition is $MRTS = w/r$ — exactly mirroring the consumer’s

$MRS = P_x/P_y$. The isoquant and isocost line are the production-theory counterparts of the indifference curve and budget line.

- **Returns to scale** determines the shape of the firm's long-run average cost curve: IRS implies falling average cost (economies of scale), CRS implies constant average cost, and DRS implies rising average cost. This drives market structure.
- **The Cobb-Douglas parameters** a and b are directly estimable from data on firms' input and output levels. They form the empirical foundation of applied production analysis.

Five Common Exam Mistakes — and How to Avoid Them

1. **Confusing diminishing marginal product with decreasing returns to scale.** Diminishing MP_L asks what happens when *one* input rises; DRS asks what happens when *all* inputs rise proportionally. A production function can have diminishing MP_L and still exhibit CRS or IRS. Always state which concept you are using.
2. **Applying the MRTS tangency condition when inputs are perfect substitutes.** For a linear production function ($Q = aK + bL$), the isoquant is a straight line and MRTS is constant. The cost-minimising firm uses only the cheaper input (corner solution) — not an interior tangency. Checking isoquant shape before applying the $MRTS = w/r$ condition is essential.
3. **Sign error on MRTS.** $MRTS_{L,K} = -\Delta K/\Delta L$ is defined as a *positive* number (the absolute value of the isoquant's slope). Moving right along an isoquant, K falls and L rises, so $\Delta K/\Delta L < 0$; the negative sign restores positivity. Do not write MRTS as a negative number.
4. **Confusing an isoquant shift with a movement along the isoquant.** A change in the input mix (substituting labour for capital) moves the firm *along* the isoquant — the isoquant itself does not shift. Only a change in *technology* shifts the isoquant inward (improvement) or outward (deterioration).
5. **Forgetting to hold the other input fixed when computing MP.** $MP_L = \partial Q/\partial L$ holds K constant. If you vary both inputs simultaneously, you are computing something else. The phrase "all other inputs held fixed" in the definition is not decorative — it is the defining restriction.

Chapter at a Glance — Week 5: Production Theory

Revision guide: everything you need before an exam or tutorial.

Key Equations

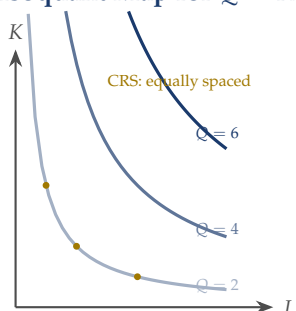
Production fn	$Q = f(K, L)$	p. 5
Total product	$TP(L) = f(\bar{K}, L)$	p. 6
MP_L	$\partial Q / \partial L _K$	p. 8
MP_K	$\partial Q / \partial K _L$	p. 8
Isoquant	$\{(K, L) : f = Q_0\}$	p. 11
MRTS	MP_L / MP_K	p. 12
RTS test	$f(\lambda K, \lambda L)$ vs λf	p. 15

Returns to Scale Diagnostic

For $Q = AK^aL^b$:

- $a + b = 1$: CRS
- $a + b > 1$: IRS
- $a + b < 1$: DRS

Isoquant Map for $Q = K^{0.5}L^{0.5}$



Consumer–Production Analogy

Consumer	Producer
$U(x, y)$	$Q = f(K, L)$
Indifference curve	Isoquant
MRS	MRTS
Dim. MRS	Dim. MRTS
IC shifts out = better	IQ shifts <i>in</i> = better
$MRS = P_x / P_y$	$MRTS = w / r$ (Wk 6)
Budget line	Isocost (Wk 6)

LO Checklist

- LO1: Consumer–production analogy ✓
- LO2: Production function; TP ✓
- LO3: MP_L , MP_K , diminishing MP ✓
- LO4: Isoquants; three special cases ✓
- LO5: MRTS; tech change vs input sub ✓
- LO6: CRS/IRS/DRS; Cobb-Douglas test ✓

Isoquant Type Guide

- **Smooth convex** → Cobb-Douglas
dim. MRTS, imperfect subs.
- **Straight line** → Linear
constant MRTS, perfect subs.
- **L-shaped** → Leontief
fixed proportions, no sub.

Key thread:

Production fn → TP → MP (one input varies) → isoquant (both vary) → MRTS (slope of IQ) → RTS (scale all inputs). Week 6 adds input prices to find the cost-minimising point.

The analogy thread: $U(x, y) \leftrightarrow Q = f(K, L)$; IC ↔ isoquant; MRS ↔ MRTS; IC shifts out ↔ IQ shifts *in*; $MRS = P_x / P_y \leftrightarrow MRTS = w / r$ (Week 6). Returns to scale asks what happens when all inputs scale — a fundamentally different question from diminishing MP.

5.10. Key Terms

Production Function

$Q = f(K, L)$: maximum output from given inputs, assuming efficient use. *Used throughout the course.*

Total Product (TP)

Output produced from a given input combination; $TP(L) = f(\bar{K}, L)$ when K is fixed.

Marginal Product (MP_L)

Additional output from one more unit of labour, all other inputs held fixed: $MP_L = \partial Q / \partial L$. *Used throughout the course.*

Law of Diminishing MP

As one input increases (others fixed), its marginal product eventually declines.

Isoquant

Curve in (L, K) space showing all input combinations producing the same output level. Production counterpart of the indifference curve. *Used throughout the course.*

MRTS

Marginal rate of technical substitution: rate at which K can be reduced per unit rise in L , output constant; $MRTS = MP_L / MP_K$.

Diminishing MRTS

MRTS falls as we move down an isoquant (more L , less K); causes isoquants to bow inward.

Input Substitution

Changing the mix of K and L while keeping output constant; movement *along* an isoquant.

Technological Change

Improvement in technology: same output achievable with fewer inputs; isoquant shifts *inward*.

Returns to Scale (RTS)

How output responds when all inputs are scaled proportionally by $\lambda > 1$.

CRS Constant returns to scale: $f(\lambda K, \lambda L) = \lambda f$. Output scales proportionally with inputs.

IRS Increasing returns to scale: $f(\lambda K, \lambda L) > \lambda f$. Output scales more than inputs. Sources: specialisation, indivisibilities.

DRS Decreasing returns to scale: $f(\lambda K, \lambda L) < \lambda f$. Output scales less than inputs. Source: coordination/management costs.

Cobb-Douglas

$Q = AK^aL^b$: CRS if $a + b = 1$, IRS if $a + b > 1$, DRS if $a + b < 1$. Most widely used production function. *Used throughout the course.*

Perfect Substitutes (production)

Linear isoquants; MRTS constant. One input can always replace the other at a fixed rate.

Leontief / Fixed Proportions

L-shaped isoquants; inputs must be used in fixed ratio. No substitution is possible.

Scale Elasticity

The percentage increase in output when all inputs rise by 1%; equals $a + b$ for Cobb-Douglas. Provides a single-number summary of returns to scale.

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter. Watch them in order for best results.

1. Total, Marginal, and Average Product (*Khan Academy*) ~ 8 min

<https://www.youtube.com/watch?v=8fm9FjDV0iA>

Covers total product, marginal product, average product, and the law of diminishing marginal product with clear diagrams. Directly reinforces Sections 5.3–5.4 and the numerical table. Ideal before Level 1–2 questions on TP and MP.

2. Marginal Rate of Technical Substitution (MRTS) (*KJ Lectures*) ~ 9 min

<https://www.youtube.com/watch?v=5mja3I52C10>

Visual walkthrough of isoquant maps, MRTS, and diminishing MRTS using the capital-labour diagram. Reinforces Sections 5.5–5.6 and is recommended before attempting the MRTS computation questions.

3. Economies and Diseconomies of Scale (*Khan Academy*) ~ 10 min

<https://www.youtube.com/watch?v=4bJs6bKjPPA>

Explains returns to scale (CRS, IRS, DRS) with clear diagrams. Directly mirrors Section 5.8. Recommended before Level 2–3 returns-to-scale questions.

4. Cost Minimisation with Isoquants and Isocosts (*Zach's Math Zone*) ~ 11 min

<https://www.youtube.com/watch?v=BltYXk4B-4c>

Walks through the cost-minimisation problem using isoquants and isocosts, showing how the optimal input bundle is found at the tangency point. Bridges production theory and cost theory, connecting Sections 5.6–5.7 to Chapter 6.

5.11. Practice Questions

How to use these questions. Arranged in three levels. Level 1 builds vocabulary and conceptual understanding; Level 2 requires algebra and diagram work; Level 3 demands synthesis across the full framework. Difficulty: ★ easy ★★ intermediate ★★★ challenging. Attempt every question independently before consulting the solutions.

Level 1: Conceptual and Short-Answer

- ★ For each consumer-theory concept, state the production-theory counterpart and explain the analogy in one sentence: (a) utility function; (b) indifference curve; (c) MRS; (d) a higher indifference curve.
- ★ Explain what it means for a production function to give the *maximum* output from given inputs. Why does actual output sometimes fall below this maximum?
- ★ A firm's total product of labour (capital fixed at $\bar{K}$) is given in the table below. Complete the MP_L column and answer the questions.

L	TP	MP_L
0	0	—
1	12	
2	22	
3	30	
4	36	
5	40	
6	42	
7	42	
8	40	

- (a) At what L does MP_L reach its maximum?
- (b) At what L does TP reach its maximum?
- (c) Why does the 8th worker *reduce* total output? Is this realistic in practice?
- (d) A rational firm would hire up to but not beyond what value of L ? Why?
- ★ In your own words, distinguish between diminishing marginal product and decreasing returns to scale. Why is confusing these two concepts one of the most common exam mistakes?
- ★ Explain, using one real-world example for each, why: (a) a software platform exhibits IRS; (b) a traditional craft brewery may exhibit DRS.
- ★ Explain the difference between input substitution and technological change. How does each appear on an isoquant diagram?

- ★ A technology consultant tells a firm: “Replace some workers with machines.” A second consultant says: “Adopt our new automated process — it produces the same output with far fewer total inputs.” Which proposal is input substitution and which is technological change?
- ★ **[Explain to a Friend — Level 1]** *No formulas allowed.* Your friend says: “Diminishing marginal product means the firm gets less and less efficient as it grows.” Write 3–4 plain-English sentences explaining why this is wrong and what diminishing marginal product actually means.
- ★ What is an isoquant? State three properties it shares with an indifference curve, and one important way in which it differs.
- ★ True or false, and explain: (a) “A firm with CRS has constant MP_L .” (b) “Two isoquants for the same firm can cross.” (c) “A technological improvement always shifts the isoquant outward.”

Level 2: Intermediate Problems

- ★★ For $Q = K^{1/3}L^{2/3}$:
 - (a) Derive MP_K and MP_L .
 - (b) Derive the MRTS.
 - (c) Show algebraically that MRTS is diminishing.
 - (d) Determine returns to scale.
 - (e) On the $Q = 8$ isoquant, find the input bundle where $MRTS = 1$. ★ *Week 6 preview: this is where $w = r$.*
- ★★ For each production function, determine CRS, IRS, or DRS. Show your working.
 - (a) $Q = 3K + 5L$
 - (b) $Q = K^{0.3}L^{0.5}$
 - (c) $Q = K^{0.5}L^{0.5}$
 - (d) $Q = K^{0.7}L^{0.8}$
 - (e) $Q = \min(2K, 3L)$ (Leontief)
- ★★ For $Q = 2K + L$ (linear / frisbee production):
 - (a) Find the MRTS. What does a constant MRTS imply about isoquant shape?
 - (b) Are K and L perfect substitutes, complements, or imperfect substitutes?
 - (c) Determine returns to scale algebraically.
 - (d) With new technology $Q = 3K + 1.5L$: for $Q = 100$, find L when $K = 10$ and when

$K = 25$. How much labour does the new technology save compared to the old?

(e) Graph the $Q = 100$ isoquants for both technologies on the same diagram.

★★ Consider the Leontief production function $Q = \min(K, 2L)$:

(a) What is the optimal ratio of K to L at any output level?

(b) Sketch the isoquant for $Q = 4$.

(c) At $(K = 6, L = 2)$, what is output? Is capital being used efficiently?

(d) Does this function exhibit CRS, IRS, or DRS? Verify.

★★ For $Q = 4K^{0.5}L^{0.5}$:

(a) Determine returns to scale.

(b) Derive the MRTS.

(c) Find the equation of the $Q = 20$ isoquant.

(d) On this isoquant find: (i) the symmetric point where $K = L$; (ii) the point where $MRTS = 2$; (iii) the point where $MRTS = 0.5$.

(e) With new technology $Q = 6K^{0.5}L^{0.5}$, find the new $Q = 20$ isoquant. Does it shift inward or outward? Sketch both.

★★ [Data-Based Question] The table records observed output levels for a firm using different combinations of capital and labour. Use it to answer the questions below.

Obs.	K	L	Q	MP_L	Notes
1	4	1	9.5	—	Baseline
2	4	2	18.0		
3	4	4	32.0		
4	4	6	42.0		
5	4	8	48.0		
6	4	10	50.0		
7	8	5	63.2		K doubled
8	8	10	89.4		K doubled

(a) Compute MP_L between consecutive observations (rows 1–6, with $K = 4$ fixed). Does MP_L diminish? At what point does it turn negative, if ever?

(b) Using rows 1 and 6: when L rises from 1 to 10 with $K = 4$ fixed, by what factor does output change? Is this consistent with diminishing returns to one input?

(c) Using rows 6 and 8: capital doubles from $K = 4$ to $K = 8$, labour stays at $L = 10$, and output rises from 50.0 to 89.4. Does this suggest CRS, IRS, or DRS? (Hint:

compare $89.4/50.0$ to $8/4 = 2$.)

- (d) Using rows 3 and 7: both inputs roughly double (from $(K = 4, L = 4)$ to $(K = 8, L = 8)$... but row 7 is $(K = 8, L = 5)$, not $(K = 8, L = 8)$). Instead, use rows 3 and 8: $(K = 4, L = 4, Q = 32)$ and $(K = 8, L = 8)$. For $Q = K^{0.5}L^{0.5}$ scaled by A , find A from row 3 and predict Q at $(K = 8, L = 8)$. Does it match?
- (e) ★ *Week 6 preview*: At observation 4 ($K = 4, L = 6$), suppose $MP_L = 5$ and $MP_K = 8$. The wage is $w = \$10$ and rental rate is $r = \$20$. Is the firm minimising cost? Compute MP_L/w and MP_K/r and explain.

★★ [Explain to a Friend — Level 2] Your friend is studying for the exam and says: “I understand that isoquants slope downward. But I don’t understand why they bow inward toward the origin rather than being straight lines. What’s the economic logic?” Write a short paragraph (4–5 sentences) explaining diminishing MRTS in plain English using a concrete capital-labour example.

Level 3: Advanced and Long-Answer Problems

★★★ For $Q = K^{0.25}L^{0.75}$:

- Derive the MRTS as a function of K and L .
- Find the MRTS at $(K = 1, L = 1)$; at $(K = 4, L = ?)$ on the $Q = 1$ isoquant; at $(K = ?, L = 4)$ on the $Q = 1$ isoquant.
- Explain what the different MRTS values in (b) mean economically.
- Determine returns to scale.
- Write the equation of the $Q = 2$ isoquant and find the symmetric point where $K = L$.

★★★ For $Q = K^{0.5}L^{0.5}$, the $Q = 6$ isoquant:

- Write the equation of the isoquant.
- Find input bundles at $K = 1$, $K = 4$, and $K = 9$. Verify all satisfy $Q = 6$.
- Compute MRTS at each point. What happens to MRTS as L increases along the isoquant?
- Technological improvement raises the function to $Q = 9K^{0.5}L^{0.5}$. Find the new $Q = 6$ isoquant. Shift inward or outward? By what proportion?
- Sketch both $Q = 6$ isoquants on a single diagram. ★ *Week 6 preview*: at what point on the original isoquant is cost minimised if $w = r$?

★★★ Returns to scale for tricky cases. For each function, determine CRS/IRS/DRS algebraically and verify numerically by doubling from $(K = 1, L = 1)$:

- $Q = (K + L)^2$

- (b) $Q = K^{0.5} + L^{0.5}$
- (c) $Q = K \cdot L$
- (d) $Q = \sqrt{KL}$
- (e) $Q = 10 + K^{0.5}L^{0.5}$ — does the standard scaling argument work when there is a constant term? Explain.

★★★ **Comprehensive consumer–production comparison.**

Part A (Consumer): $U(x, y) = \ln(x) + \ln(y)$, income m , prices P_x, P_y .

- (a) Find the optimal bundle (x^*, y^*) .
- (b) What income share goes to each good?
- (c) If P_x rises by 10% (others unchanged), what happens to optimal x and y ?
- (d) Separately identify the substitution and income effects on x .

Part B (Production): $Q = K^\alpha L^\beta, \alpha + \beta = 1$ (CRS).

- (e) Derive the MRTS.
- (f) Show that MRTS depends only on the ratio K/L along any isoquant.
- (g) If the firm doubles only L (holding K fixed), does output double? Show algebraically.
- (h) If the firm doubles both K and L , does output double? Explain why the answers to (g) and (h) differ.

★★★ **Cobb-Douglas: proof and numerics.** For $Q = AK^aL^b$:

- (a) Show algebraically that $Q' = \lambda^{a+b}Q$ when all inputs scale by λ .
- (b) State the CRS, IRS, and DRS conditions in terms of a and b .
- (c) For $Q = 2K^{0.4}L^{0.4}$: determine returns to scale; derive MP_L and show it is diminishing.
- (d) For $Q = K^{0.6}L^{0.6}$: start at $(1, 1)$, then double to $(2, 2)$. By what factor does output change? Consistent with IRS?
- (e) Prove that for Cobb-Douglas, a and b are the output elasticities with respect to K and L respectively. (Output elasticity $= \partial \ln Q / \partial \ln K$.)

★★★ **Three perspectives on $Q = K^{0.5}L^{0.5}$ at $(K = 4, L = 4)$:**

- (a) **Intuition:** Describe in plain English what the MRTS tells you at this symmetric point. What does it mean that $MRTS = 1$ here?
- (b) **Algebra:** Derive MP_L, MP_K , and MRTS at $(K = 4, L = 4)$. Verify $MRTS = 1$.

- (c) **Policy:** A subsidy reduces the effective cost of capital by 50% (from r to $0.5r$). The wage is unchanged. Using the cost-minimisation condition $MRTS = w/r$ (Week 6 preview), in which direction will the firm adjust its input mix? More capital-intensive or more labour-intensive?

★★★ [Explain to a Friend — Level 3] Your friend is a business student studying firm strategy. She has heard that “firms should always try to achieve constant returns to scale.” Write a careful 5–6 sentence explanation addressing: (a) why IRS is actually *desirable* from the firm’s perspective (lower average cost); (b) why a market with IRS tends toward concentration or monopoly; and (c) one real-world industry where DRS is observed and why being large is not always an advantage.

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.